Sidewalk Navigation Assistance Device (SNAD)

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Background

- According to an analysis of the 1997-2010 National Health Interview Survey, 518,000 fall injuries occurred annually on outdoor streets and sidewalks [1].
- This exemplifies the need to improve mobility and accessibility for individuals with visual impairments, seniors, and others who face challenges in navigating walkways.
- Sidewalks often have hazards like uneven surfaces, cracks, bumps, and obstacles, making it difficult for these individuals to move safely and independently.
- To reduce the frequency of trip-related injuries on sidewalks and walkways, we propose a personal device to accurately detect sidewalk cracks and effectively alert the user to these hazards.

Project Goal

- This project aims to develop a device capable of using a machine learning model to perform real-time image classification for the purpose of reducing trip-related injuries on sidewalks and walkways.
- The target demographic for this device is primarily the elderly population due to commonly occurring difficulties with vision, foot clearance, and general autonomy.

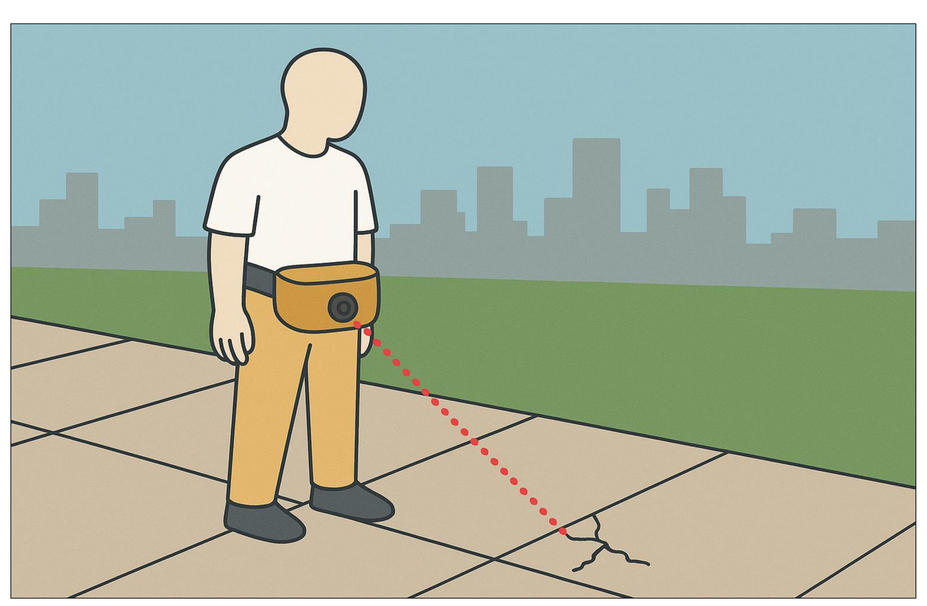

Figure 1. Illustrative Use Case Example

Materials

- Raspberry Pi 4 Model B
  - Raspberry Pi is a small and affordable single-board computer capable of performing powerful computations.
- HiLetgo OV5647 5MP Camera
  - A compact and low-cost camera capable of 1080p video resolution.
- 3-D Printed Custom Designed Camera Mount
- Soberton Inc. WST-1210T Passive Buzzer
  - Sound Pressure Level (SPL) of 85dB
- ROMOSS Solo5 10000mAh Power Bank
  - Provides up to three continuous hours of use time.

Design Details

- The SNAD is designed to detect sidewalk height offsets greater than 10mm; this value is determined as a reasonable compromise where the height is still distinguishable enough to be reliably classified by our device, but where the offset is still small enough that it does not include trivial hazards.
- The device is packaged and contained within a lightweight fabric pouch that can be worn comfortably on the waist, providing light dust and water resistance for the electronic components.
- After powering on the device, the live image classification will begin immediately; if a hazard is detected, the user will be promptly alerted with a high-pitched beep, warning the individual to pay close attention to their surroundings.

Methods

- Sidewalk training image dataset was captured using a Raspberry Pi 4 oriented at hip level.
- From an initial set of 40,000 Hazard/Non-Hazard images, a curated subset of ~6,000—captured under varying times and weather conditions—led to significant improvements in model performance and generalizability.
- Images were then resized to 480x260 pixels, converted to grayscale, and pixel values were normalized with mean ($\mu = 0.4$) and standard deviation ($\sigma = 0.6$).
- Random horizontal flip transformation with probability (p=0.75) was applied
- Convolutional Neural Network (CNN) with following architecture:
  - Four Convolutional Layers (kernel size = 3x3)
  - One Dropout Layer (p =0.5)
  - Four ReLu Activation Layers
  - Four Pooling Layers, using a combination of max and average pooling

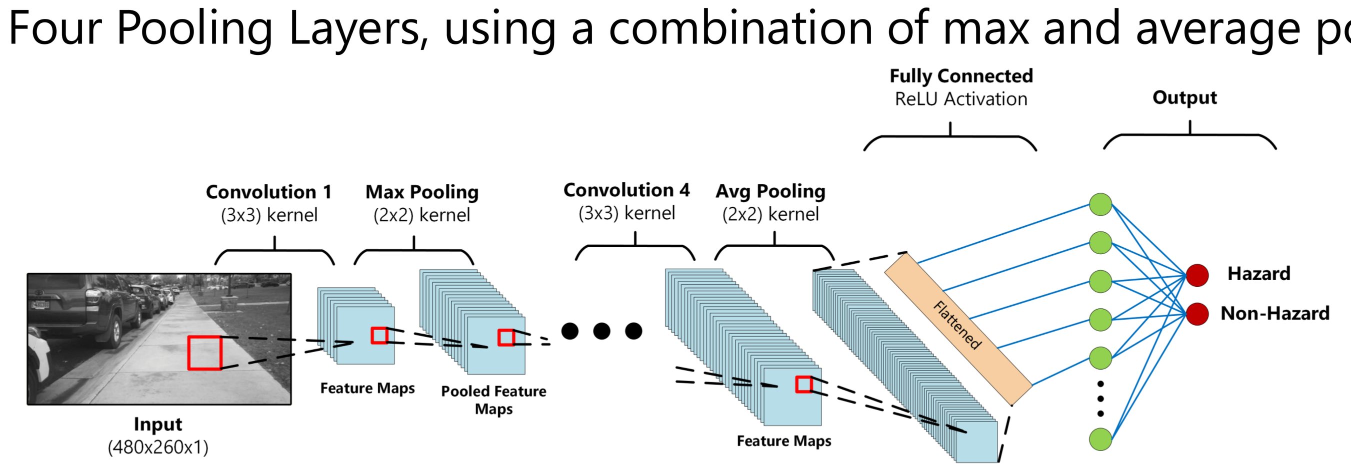

Figure 2. CNN Model Architecture

- The dataset split into 80% training, 10% validation, and 10% testing sets.
- The model implemented using PyTorch with a binary-cross-entropy loss function and Adam optimizer (learning rate *lr = 0.001*).
- The model was trained for 15 epochs with a batch size of 32.
- Training and validation loss are shown to decrease steadily, demonstrating effective learning, while training and validation accuracy increase from initially ~50% to nearly 90%, indicating that the model is able to apply what has been learned to new data. This is confirmed by a testing accuracy of 87%.

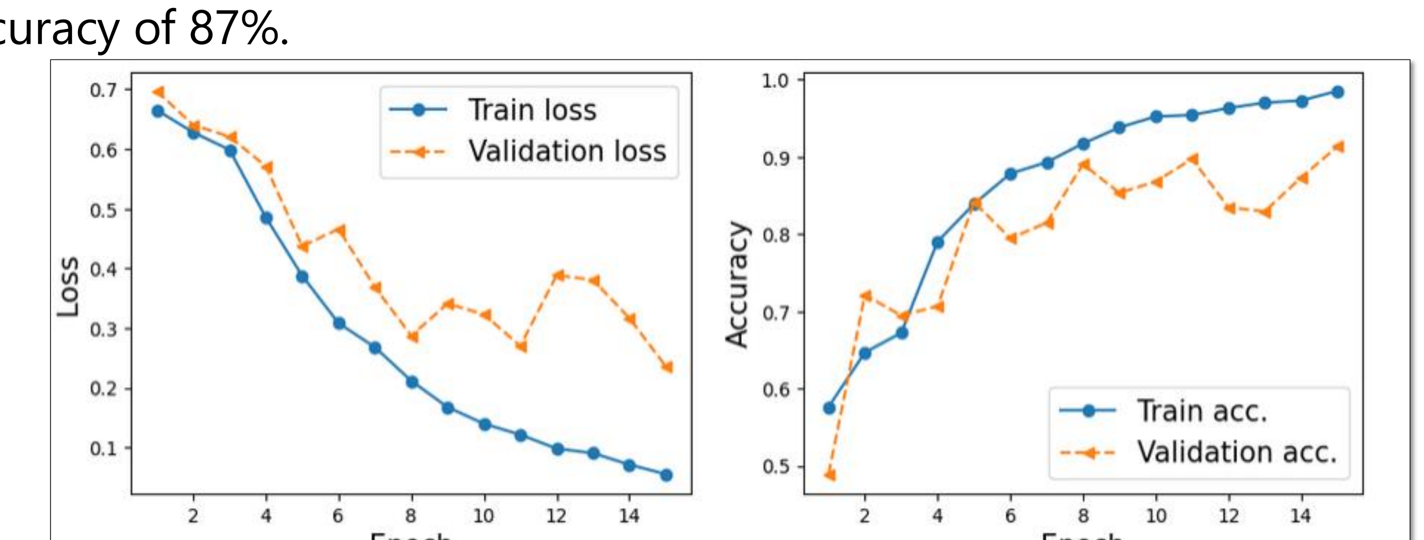

Figure 3. Model Training Loss & Accuracy vs. Epochs

Results

- SNAD was deployed using a Python script that captured images, processed them as inputs to the CNN model, and activated a buzzer based on the classification output.
- The device processed images at a rate of approximately two frames per second, which is sufficient for real time hazard detection in the context of the user.
- To further improve accuracy, a buzzer mechanism was introduced to mitigate the effects of misclassifications. Buzzer only activates if 3 of the past 5 model predictions are classified as hazards.

Actual \ Predicted	Hazard	Non-Hazard
Hazard	104 (True Positive)	3 (False Negative)
Non-Hazard	30 (False Positive)	~2500 (True Negative)

Figure 4. Confusion Matrix from Live Testing

Figure 5. Live Image Classification Results and Probabilities

Future Work

- To improve the performance of the SNAD, future implementation may include functionality to respond to the distance between the hazard and the user.
- Methods to improve the warning system by adjusting the frequency of the buzzer to the proximity of the hazard may also be implemented.
- The system could be migrated to the Raspberry Pi 5 for faster live inference performance. Additionally, the model could be run on a Raspberry Pi 5 AI hat for maximum performance.